

Global transport in soaring flights

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Chapter 3

What motion should a stochastic model reproduce?

$$\mathbf{X}_f(t, \tau) = \mathbf{r}_f(t + \tau) - \mathbf{r}_f(t), \quad R_f = |\mathbf{X}_f|.$$

$$M_2^{\text{flight}}(\tau) = \frac{1}{N_f(\tau)} \sum_f \underbrace{\frac{1}{n_f(\tau)} \sum_k R_{fk}^2}_{\text{mean within flight } f}.$$

Windows stay within a retained segment. The main comparison covers $10 \leq \tau \leq 10^4$ s.

Displacement growth, distribution shape, directional preference and memory must be confronted together.

Retained and eligible populations

Archive: 156,406 paragliders and 6,094 hang gliders.

Full diagnostics: 155,085 and 6,060 eligible flights, all supported segments, on a 10-s grid.

Archive TAMSD: equal segment weights. Full V_1, V_2 : equal flight weights. Baseline quantiles and moments: pooled windows.

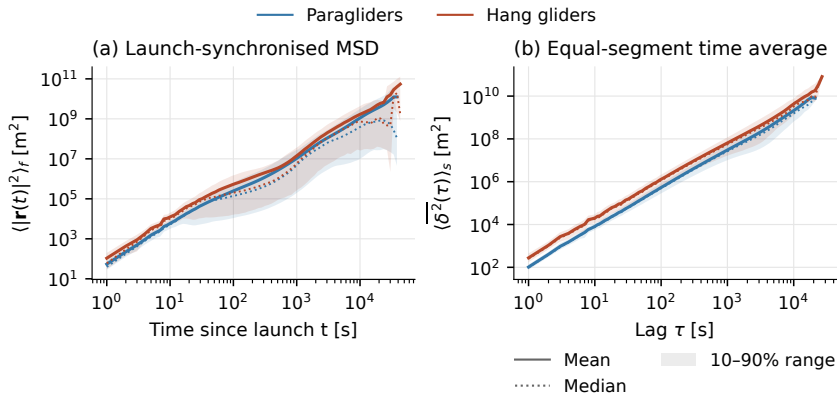
Launch, segment and flight averages use different weights

Estimator	Unit receiving equal weight	Support requirement
Launch ensemble	available flights at elapsed time t	observed coordinate at that time
Archive TAMSD	eligible retained segments	native-grid displacement within segment
Full V_1, V_2	eligible flights at lag τ	within-segment common-grid stencil
Pooled moments	available increment origins	stated non-overlapping baseline stencil

Different exponents can reflect different weights and changing support. Their difference alone is not an ergodicity test.

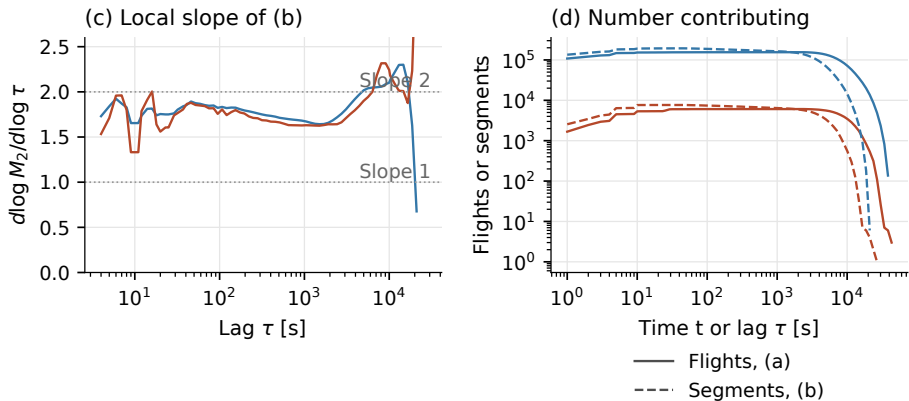
Which estimator corresponds to the physical population the model should reproduce?

Launch averages and time averages use different populations



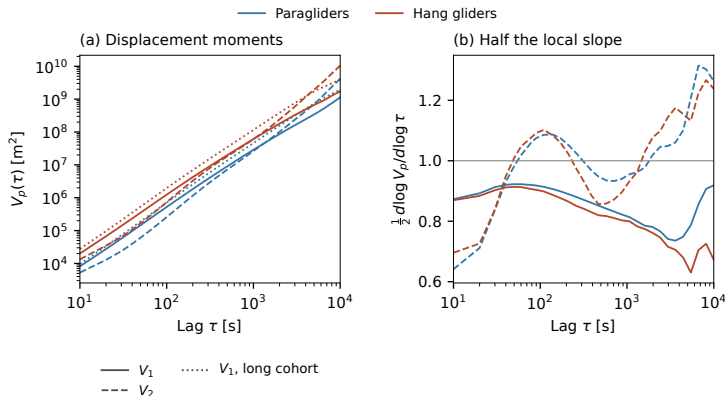
How much of the difference follows launch synchronization and segment weighting?

Local slopes must be read together with falling support



Which scale interval supports a stable finite-range description?

The MSD grows faster than linearly on the measured interval



	b_1	b_2
Para	1.703	2.003
Hang	1.649	1.985

$V_1 = \langle R^2 \rangle$; V_2 uses second differences.

The local slope bends. One fitted number does not establish a power law.

Full eligible archive; 10–10,000 s. Solid: V_1 ; dashed: V_2 ; dotted: V_1 , fixed long cohort. Right: half the local slope. Descriptive fits; no calibrated confidence intervals.

Constant velocity cancels; an intrinsic H remains open

$$A_2 \mathbf{r} = \mathbf{r}(t + 2\tau) - 2\mathbf{r}(t + \tau) + \mathbf{r}(t), \quad V_2 = \langle |A_2 \mathbf{r}|^2 \rangle.$$

Exact cancellation

$$\mathbf{r}(t) = \mathbf{r}_0 + \mathbf{v}_d t + \mathbf{Y}(t) \quad \Rightarrow \quad A_2 \mathbf{r} = A_2 \mathbf{Y}.$$

The constant velocity may combine wind, route choice and net progress. A turn or spatially varying wind survives.

Conditional interpretation

If centred increments have finite variance, stationary statistics and the law $K\tau^{2H}$,

$$V_2 = K(4 - 2^{2H})\tau^{2H}, \quad 0 < H < 1.$$

At $H = 1$ the prefactor vanishes. Our $b_2 \simeq 2$ cannot be read as a validated $H \simeq 1$.

Which residual and which scale interval could satisfy the assumptions needed to estimate H ?

At $\tau = 10^4$ s, V_2 retains only 14,360 para / 563 hang flights; it needs 2τ of continuous support.

Second differences measure changes of interval-mean velocity

$$\bar{\mathbf{v}}_1 = \frac{\mathbf{r}(t + \tau) - \mathbf{r}(t)}{\tau}, \quad \bar{\mathbf{v}}_2 = \frac{\mathbf{r}(t + 2\tau) - \mathbf{r}(t + \tau)}{\tau}.$$

$$\frac{A_2 \mathbf{r}}{\tau} = \bar{\mathbf{v}}_2 - \bar{\mathbf{v}}_1, \quad D_2(\tau) = \frac{\sqrt{V_2(\tau)}}{\tau} = \sqrt{\langle |\bar{\mathbf{v}}_2 - \bar{\mathbf{v}}_1|^2 \rangle}.$$

D_2 is in m/s. At $\tau = 60$ s it compares the mean velocity vector in one minute with the vector in the next minute. Changes of magnitude and direction both contribute.

At which scales does the ground motion change most, and how does that depend on the region?

Exact kinematic identity; no power law or stationarity assumption is needed.

Why a slope near two need not identify a Hurst exponent

$$V_2(\tau) = \tau^2 D_2^2(\tau), \quad b_{2,\text{loc}} = 2 + 2 \frac{d \log D_2}{d \log \tau}.$$

Saved full-archive curves

Over 10–10,000 s, D_2 varies within:

	D_2 range (m/s)
Paragliders	4.9–7.3
Hang gliders	7.6–11.5

A nonmonotone change of less than a factor two across three decades of lag.

Interpretation and limits

The τ^2 factor can dominate growth; local curvature prevents one validated H .

The fixed long-flight cohort also gives $b_2 > 2$: flight membership alone cannot explain it. Origins and weights still change.

Does a stable slope survive with flights and origins fixed, within comparable flight phases?

Thesis Sec. 3.1.2; saved full-archive variations, unchanged numerical run. Ranges are descriptive, not confidence intervals.

Matteo Di Sante | Chapter 3: global transport

Finite persistence can produce a crossover without nonstationarity

A stationary centred velocity with variance σ_v^2 and normalised correlation $C(s) = e^{-s/t_p}$ gives

$$V_2(\tau) = 2\sigma_v^2 t_p^2 \left[2\frac{\tau}{t_p} - 3 + 4e^{-\tau/t_p} - e^{-2\tau/t_p} \right].$$

$$\tau \ll t_p : \quad V_2 \sim \frac{4\sigma_v^2}{3t_p} \tau^3, \quad \tau \gg t_p : \quad V_2 \sim 4\sigma_v^2 t_p \tau.$$

A local slope near two occurs between these regimes despite stationary position increments.
[This is an analytical example, not a fit to the flights.](#)

In flights, turns, glide-climb transitions and changing wind survive constant-velocity cancellation. Their characteristic times, combined with changing observation support, are a plausible origin of the measured curvature.

Which dominates the bends: turning, phase changes or sampling? Can phase-conditioned and common-origin comparisons separate them?

Thesis Sec. 3.1.2 and the velocity-covariance model in Sec. 3.6; the closed formula was checked against covariance integration.

Third differences also cancel a constant acceleration

$$A_3 \mathbf{r} = \mathbf{r}(t + 3\tau) - 3\mathbf{r}(t + 2\tau) + 3\mathbf{r}(t + \tau) - \mathbf{r}(t), \quad V_3 = \langle |A_3 \mathbf{r}|^2 \rangle.$$

For $\mathbf{r} = \mathbf{r}_0 + \mathbf{v}_0 t + \frac{1}{2} \mathbf{a} t^2$ with constant $\mathbf{a}$,

$$V_2 = |\mathbf{a}|^2 \tau^4, \quad V_3 = 0.$$

Under the centred stationary-increment power law,

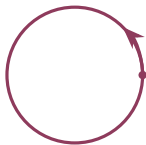
$$\frac{V_3}{V_2} = \frac{15 - 6 \cdot 2^{2H} + 3^{2H}}{4 - 2^{2H}}, \quad 0 < H < 1.$$

Do the same supported observations satisfy the predicted relation between orders?

Order three requires 3τ of continuous support; the model law must hold through 3τ .

Open and closed tasks change the growth of displacements

Open straight course



Closed circular course

Full eligible archive

	Task	N	b_1	b_2
Para	Open	67003	1.770	1.937
	Closed	87807	1.623	2.036
Hang	Open	2183	1.688	1.892
	Closed	3877	1.628	2.007

The ordering reverses at order two. Constant-drift cancellation leaves the return geometry in the observable.

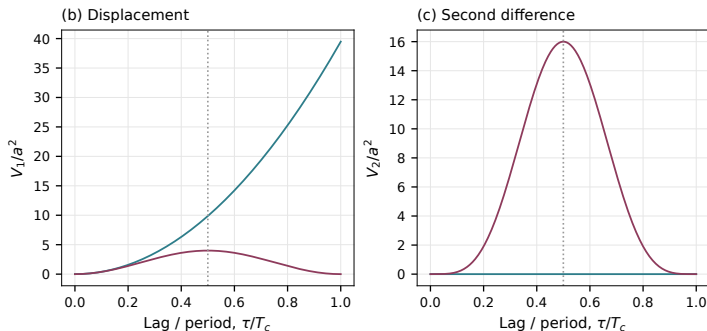
Compare tasks at matched duration and region, using both physical lag and τ/T .

Task declarations; descriptive slopes, 10–10,000 s. Drawing is a geometric illustration. [Exact circular example](#).

A circle retains its geometry after second differencing

For $\mathbf{r}(t) = a(\cos \omega t, \sin \omega t)$ and $T_c = 2\pi/\omega$,

$$V_1 = 4a^2 \sin^2(\pi\tau/T_c), \quad V_2 = 16a^2 \sin^4(\pi\tau/T_c).$$



Which scale effects could follow return geometry without a change of stochastic law?

Observed-path closure needs special care for segmented flights

$$c_f^{\text{obs}} = \frac{|\mathbf{r}_{f,\text{last}} - \mathbf{r}_{f,\text{first}}|}{\sum_{s \in \mathcal{S}_f} \sum_k |\mathbf{r}_{fs,k+1} - \mathbf{r}_{fs,k}|}.$$

The denominator omits unobserved intervals. For a segmented flight, c_f^{obs} need not be bounded by one.

Declared task, observed endpoint separation and actual flown-route closure are related descriptions with different information requirements.

Does the task contrast persist when restricted to continuously observed flights?

A single displacement scale predicts equal quantile slopes

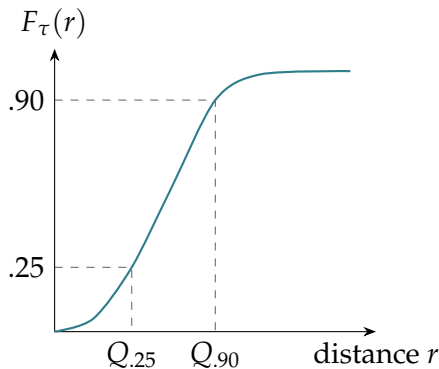
$$Q_p(\tau) = \inf\{r : F_\tau(r) \geq p\}.$$

An H -self-similar process with stationary increments predicts

$$R(\tau) \stackrel{d}{=} (\tau/\tau_0)^H R(\tau_0),$$

$$Q_p(\tau) = (\tau/\tau_0)^H Q_p(\tau_0).$$

$H_p = H$ for every p ; $Q_{.90}/Q_{.25}$ stays fixed.



Schematic CDF. Quantiles describe ranks within one distribution; they do not define permanent classes of slow and fast flights. Equal slopes are necessary, not sufficient, for process self-similarity.

The full archive and the fixed cohort answer different questions

Population	Paragliders	Hang gliders
Retained cleaned flights	156,406	6,094
Eligible for the common-grid diagnostic	155,085	6,060
Fixed long-flight control	14,360	563

Every eligible segment contributes where it supports the lag. The long-flight control adds a common population and common origins across the complete lag range.

Which statement concerns the observed archive, and which is conditional on long flights?

The fixed control requires common temporal support

All eligible flights contribute at each supported lag in the archive-wide analysis. The additional fixed cohort contains every flight with a retained segment lasting at least 20,000 s.

This provides at least 1001 common ten-second origins in one qualifying segment at the largest lag of 10,000 s. Shorter segments of those flights contribute wherever they support the same full lag stencil.

How different are the shorter-flight results on a shorter common lag interval?

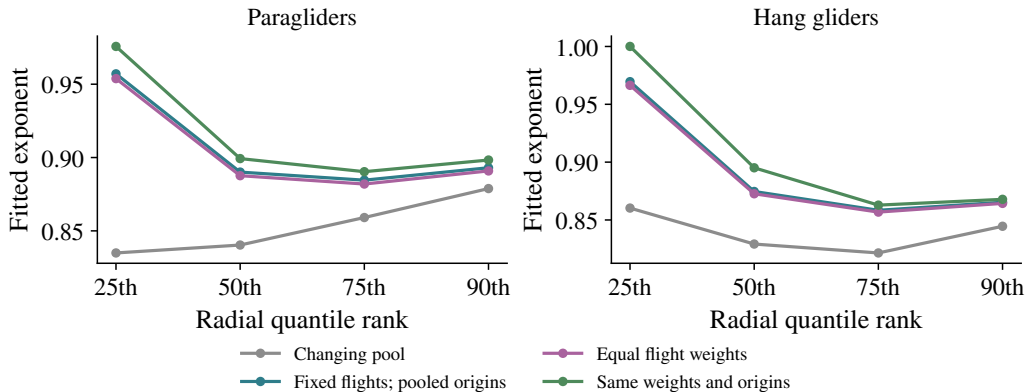
Holding flights fixed does not by itself hold the observations fixed

1. All available flights and origins, pooled.
2. Fixed flights, still pooling their available origins.
3. Fixed flights with equal total flight weight.
4. Fixed flights, equal flight weight and the same origins at every lag.

$$F_{\tau}(r) = \frac{1}{N_*} \sum_f \frac{1}{n_f^*} \sum_{k \in \mathcal{O}_f^*} \mathbf{1}\{R_{fk}(\tau) \leq r\}.$$

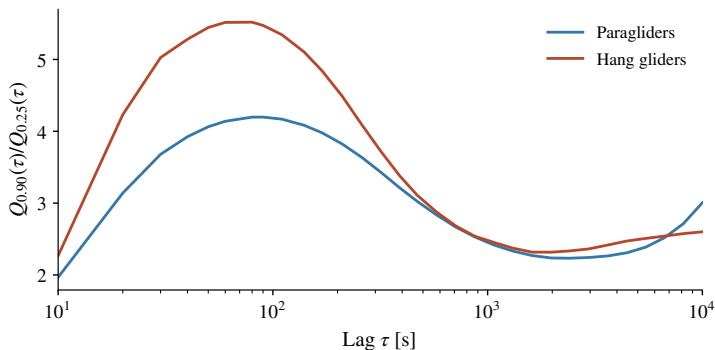
How much of a change in the quantile curve comes from membership, weights or origin selection?

Membership, flight weights and common origins change the comparison



Which part of the slope change comes from each population control?

Relative spread changes, but the change is not monotone



$$\frac{d \log(Q_{.90}/Q_{.25})}{d \log \tau} = H_{.90}(\tau) - H_{.25}(\tau).$$

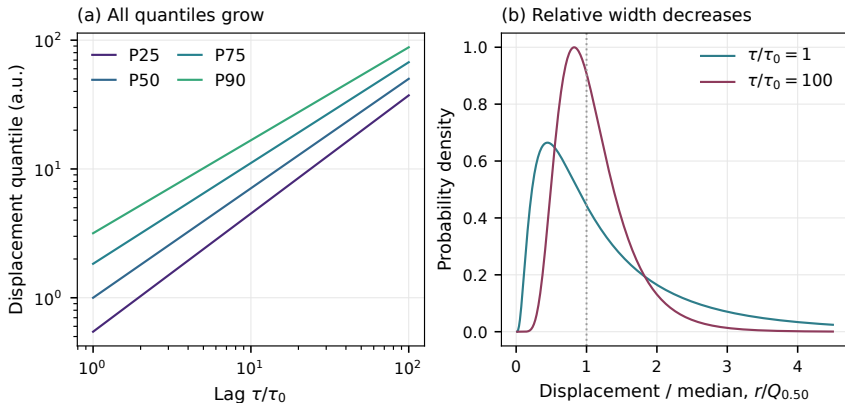
The ratio first rises, then falls, then rises again.

A negative fitted slope summarizes part of that evolution; it does not imply narrowing at every lag.

How much of this shape evolution survives site-day uncertainty, route matching and phase conditioning?

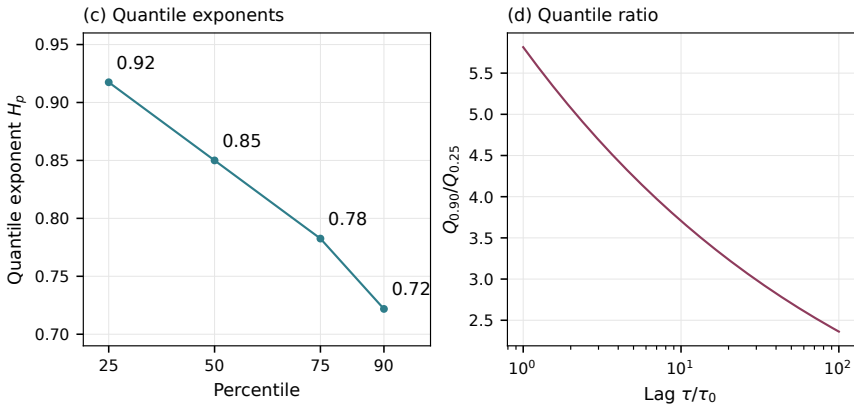
Fully controlled population. A lower ratio means less relative spread by this measure; it does not establish a flatter density tail.

All quantiles can grow while spread relative to the median narrows



How can a changing distribution shape coexist with increasing displacement scales?

Unequal rank exponents can encode that changing shape



Which empirical observations would distinguish this example from a true common scale family?

Anisotropy does not prevent a common scalar rescaling

$$\mathbf{X}(\tau) \stackrel{d}{=} \tau^H \mathbf{Z} \implies |\mathbf{X}(\tau)| \stackrel{d}{=} \tau^H |\mathbf{Z}|,$$

even if the law of $\mathbf{Z}$ is anisotropic.

With two centred independent Gaussian components instead,

$$\mathbb{E}[R^2] = a\tau^{2H_E} + b\tau^{2H_N},$$

unequal component exponents can produce a lag-dependent radial effective exponent.

Is a mismatch caused by unequal directional scales, changing marginal shape or changing joint structure?

East, north and radius together still do not identify the joint law

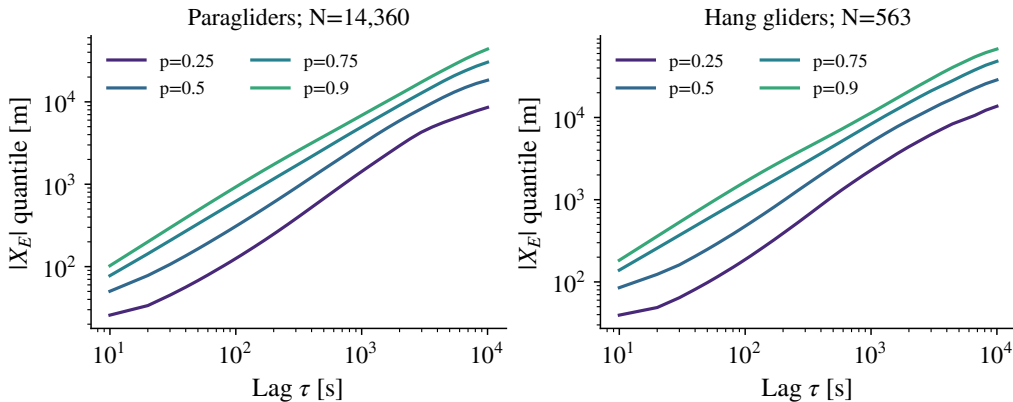
$$\{(1,2), (-1,-2)\} \quad \text{versus} \quad \{(1,-2), (-1,2)\},$$

each pair equiprobable.

Quantity	First law	Second law
Signed east marginal	± 1	± 1
Signed north marginal	± 2	± 2
Radius	$\sqrt{5}$	$\sqrt{5}$
Cross-covariance	$+2$	-2

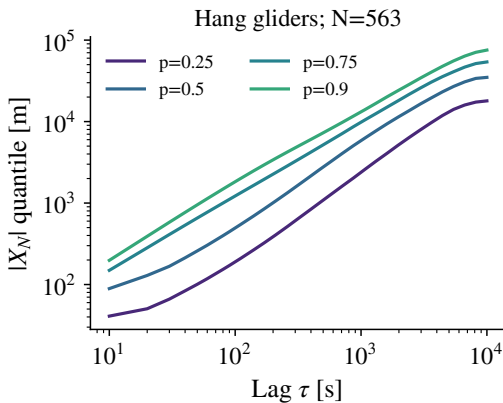
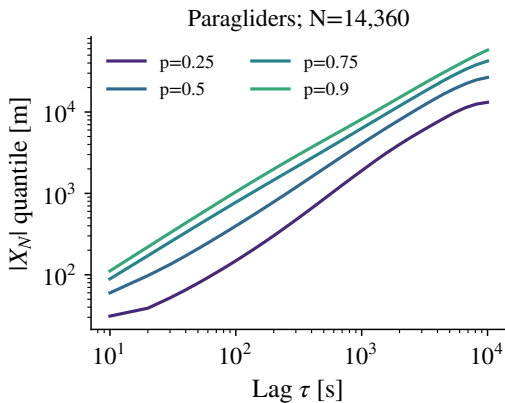
The missing information is real: signs and component dependence require a joint diagnostic.

Fixed-population quantiles: east component



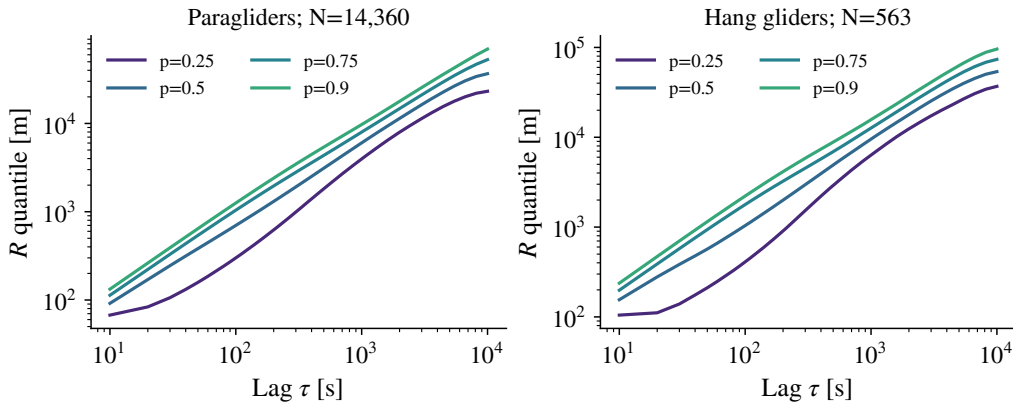
Do the ranks retain one relative shape as the lag increases?

Fixed-population quantiles: north component



Do the ranks retain one relative shape as the lag increases?

Fixed-population quantiles: radial displacement



Do the ranks retain one relative shape as the lag increases?

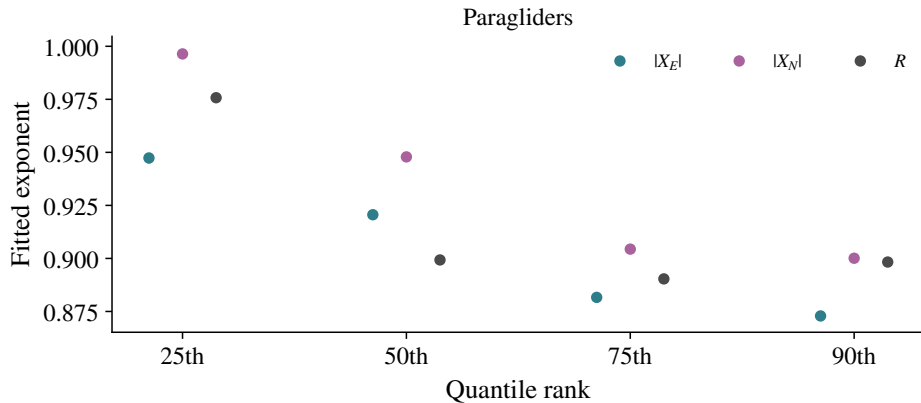
Whole-flight bootstrap preserves pairing across ranks, axes and lags

Resample flights with replacement 400 times. Each selected flight retains all its common origins and its joint east–north observations at every lag.

Overlapping origins remain within the resampled flight; they are not treated as independent replicates.

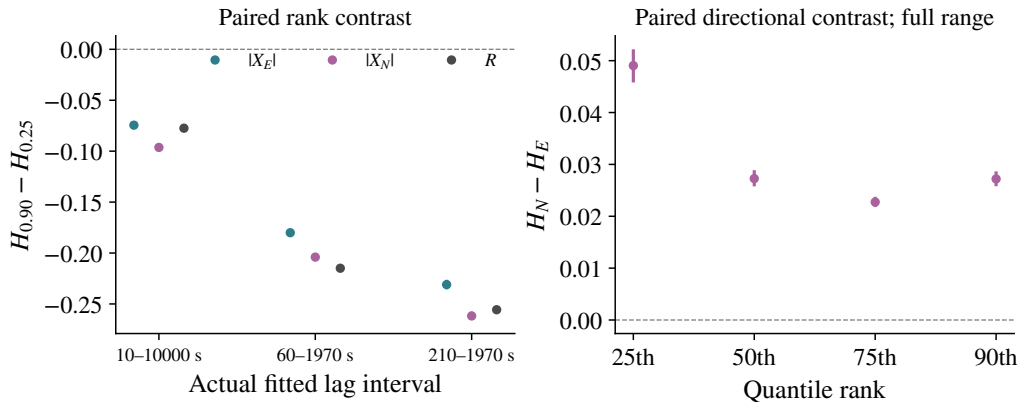
How would shared pilot history or site–day weather change these uncertainty estimates?

Paragliders: component and radial exponent estimates



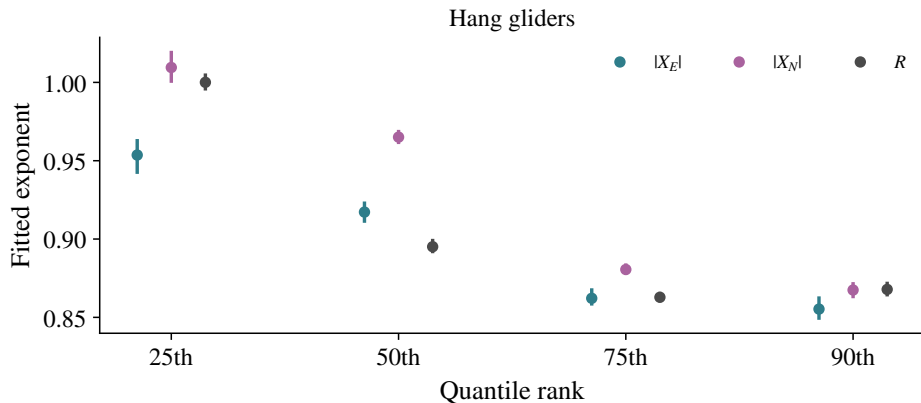
Which differences need a paired contrast instead of comparing overlapping intervals?

Paragliders: paired rank and directional contrasts



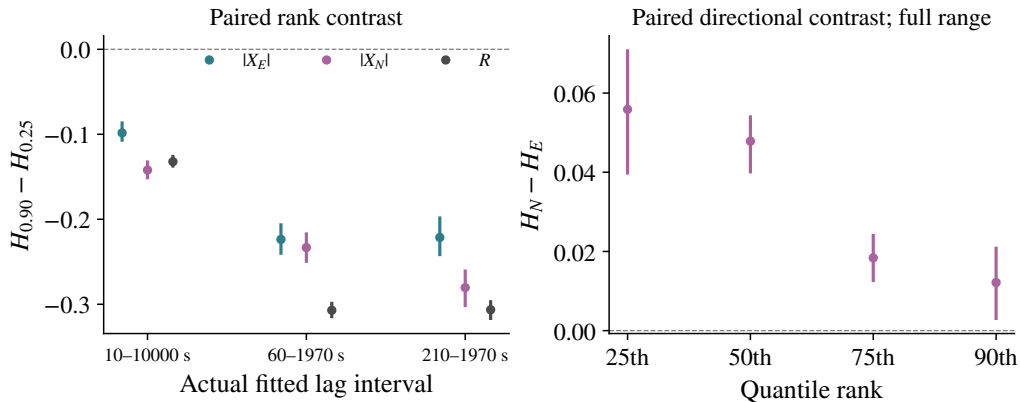
Which differences persist under a change of fitting interval?

Hang gliders: component and radial exponent estimates



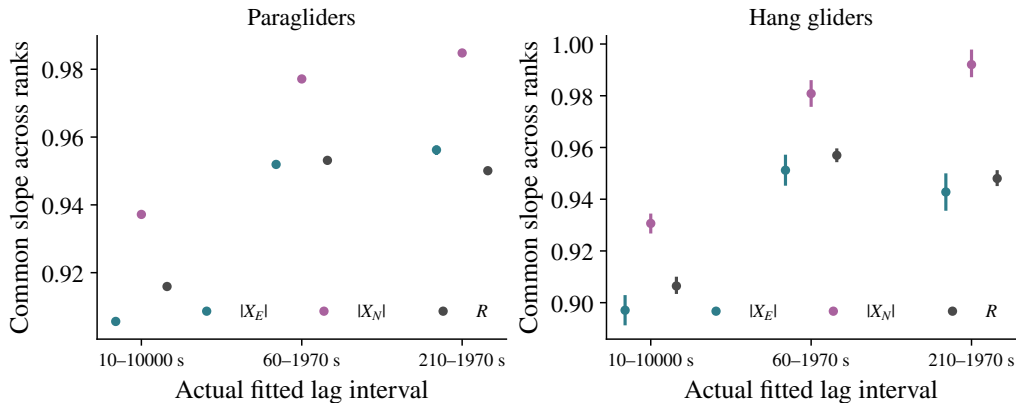
Which differences need a paired contrast instead of comparing overlapping intervals?

Hang gliders: paired rank and directional contrasts



Which differences persist under a change of fitting interval?

A common fitted scale depends on the fitted interval



Which range has an independent physical justification?

A large density near zero need not carry much probability

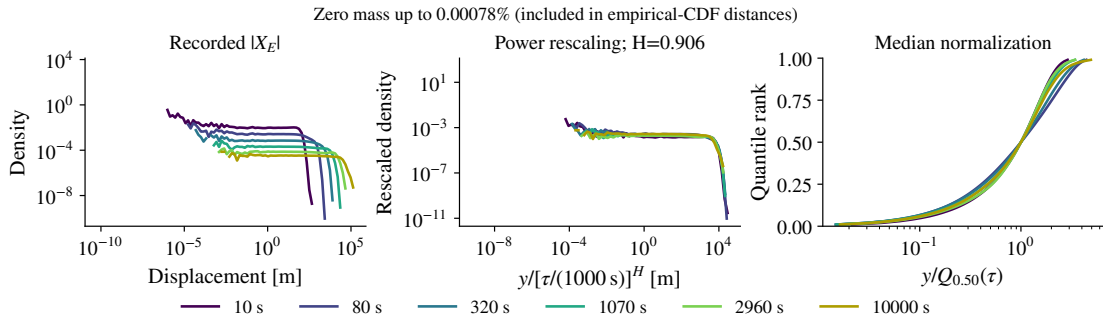
$$\hat{f}_j = \frac{p_j}{b_{j+1} - b_j}.$$

A very narrow histogram bin can have a tall density and still contain little probability. Across the displayed marginal laws, the mass below 1 mm, including zero and the straddling bin, is below 0.003%.

That numerical scale is not a claim of millimetre GNSS resolution. The following plots retain positive support; empirical-CDF comparisons also retain zero mass.

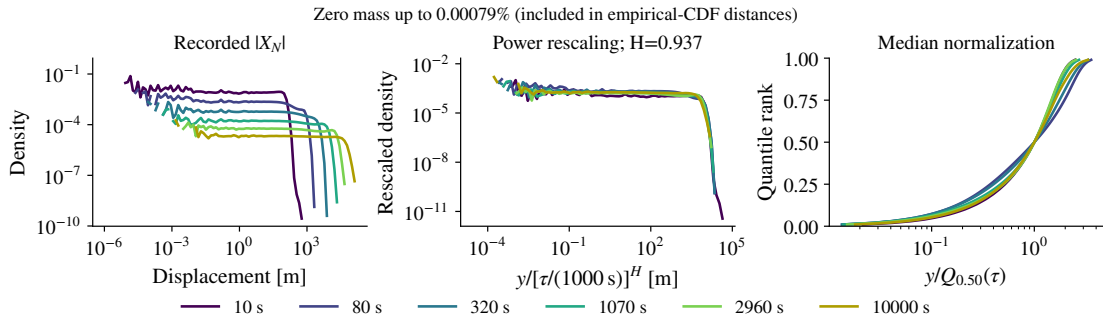
Do the central ranks and the probability-bearing tails tell the same scaling story?

Paragliders: marginal collapse for east



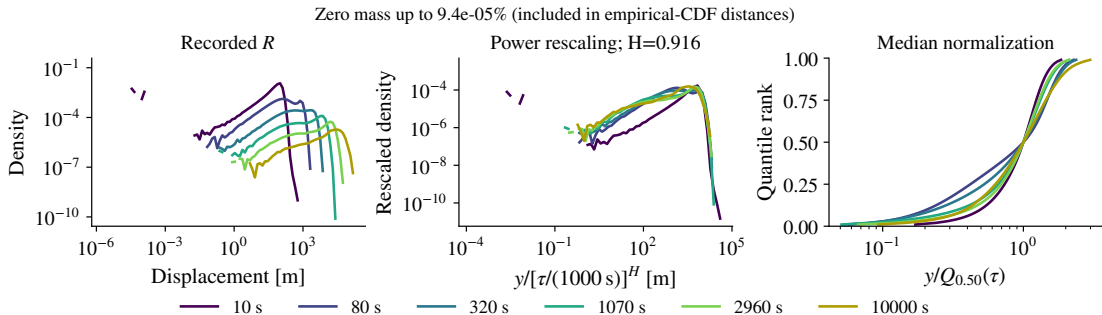
Does power or median normalization leave a systematic shape change?

Paragliders: marginal collapse for north



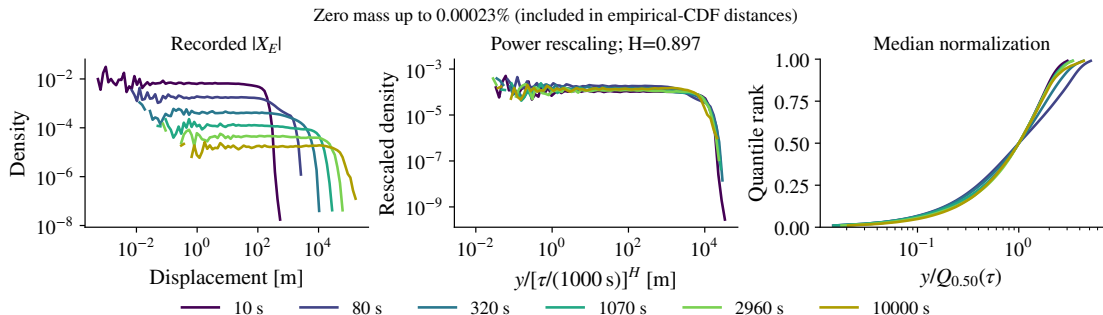
Does power or median normalization leave a systematic shape change?

Paragliders: marginal collapse for radius



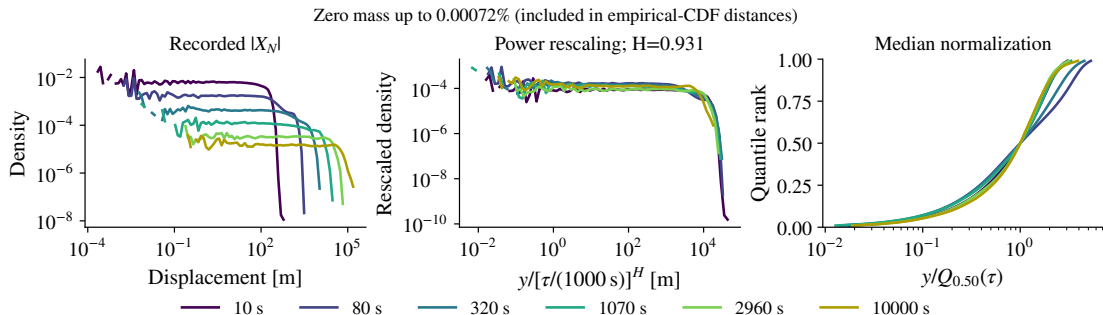
Does power or median normalization leave a systematic shape change?

Hang gliders: marginal collapse for east



Does power or median normalization leave a systematic shape change?

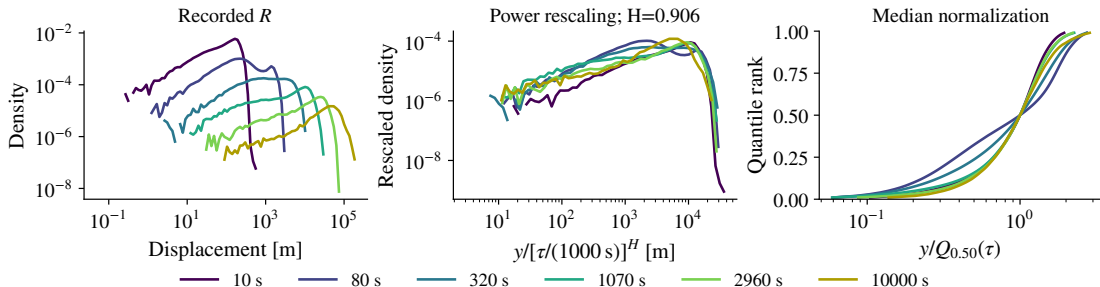
Hang gliders: marginal collapse for north



Does power or median normalization leave a systematic shape change?

Hang gliders: marginal collapse for radius

Zero mass up to 0.00018% (included in empirical-CDF distances)



Does power or median normalization leave a systematic shape change?

The joint comparison keeps geographic axes and one scalar scale

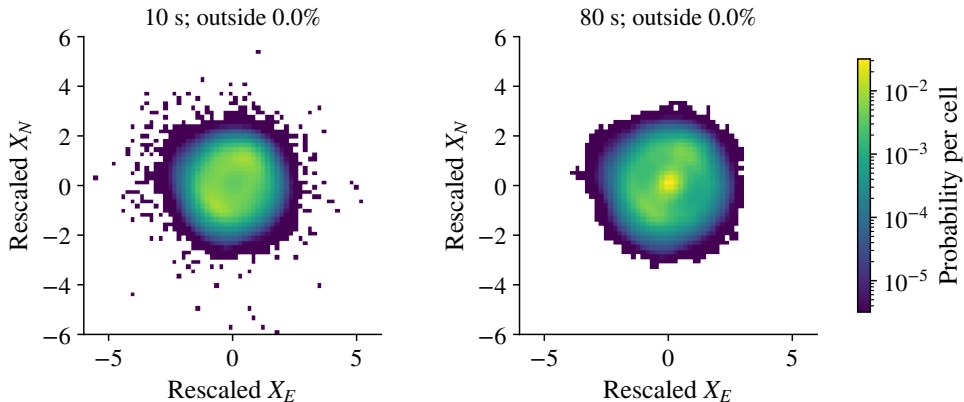
$$\mathbf{Z}_\tau = \frac{(X_E, X_N)}{s_0 \left(\frac{\tau}{1000\text{s}} \right)^{H_*}}.$$

H_* is the common fit across all component/rank slopes. The radial reference median near 1000 s is adjusted to 1000 s by the same power.

There is no lag-specific centring, rotation or whitening. The six lags share the same cells and equal total weight per flight.

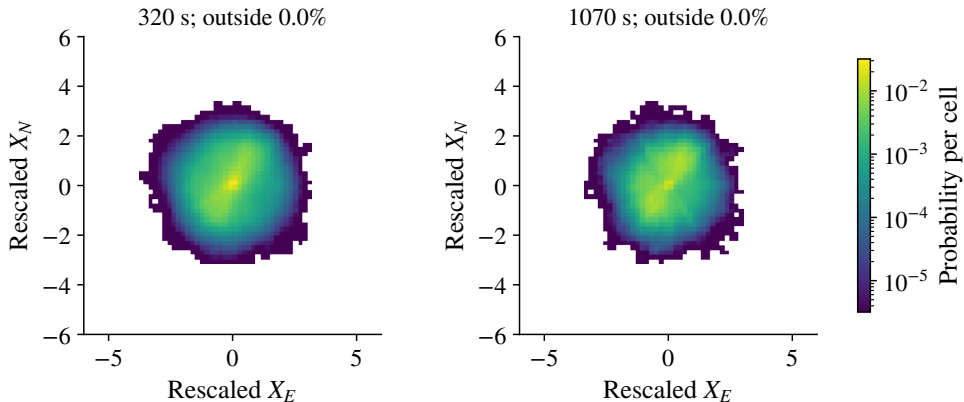
Which directional and dependence changes remain visible when the normalization cannot remove them separately?

Paragliders: signed joint laws (1/3)



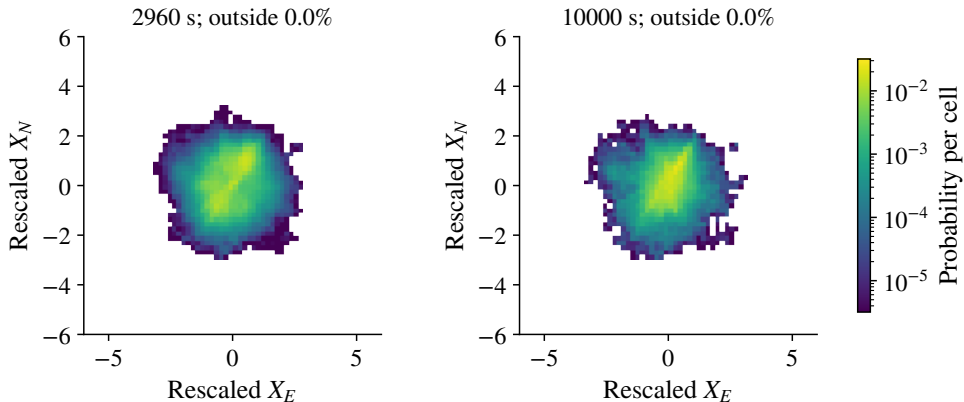
Which changes concern orientation, central concentration or mass outside the displayed square?

Paragliders: signed joint laws (2/3)



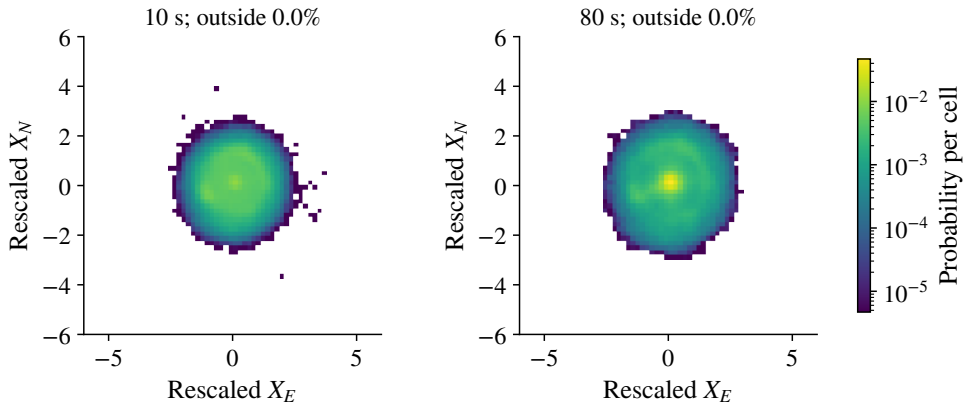
Which changes concern orientation, central concentration or mass outside the displayed square?

Paragliders: signed joint laws (3/3)



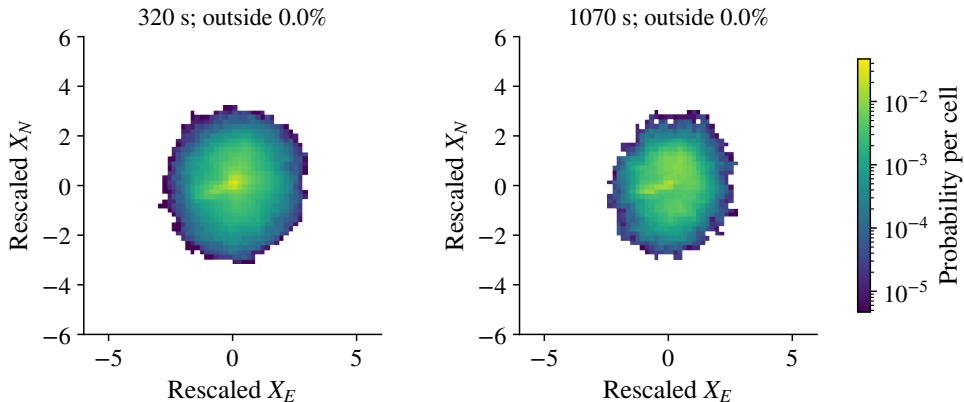
Which changes concern orientation, central concentration or mass outside the displayed square?

Hang gliders: signed joint laws (1/3)



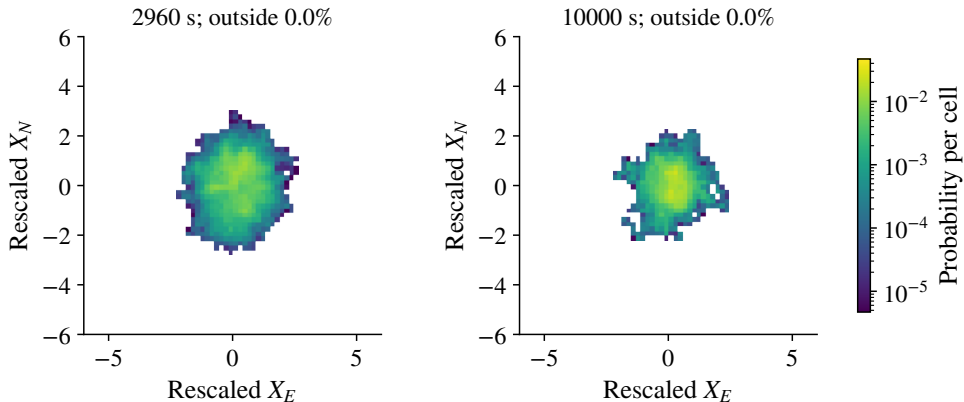
Which changes concern orientation, central concentration or mass outside the displayed square?

Hang gliders: signed joint laws (2/3)



Which changes concern orientation, central concentration or mass outside the displayed square?

Hang gliders: signed joint laws (3/3)



Which changes concern orientation, central concentration or mass outside the displayed square?

One scalar rescaling leaves changes in the signed joint law

	Paragliders	Hang gliders
Common scalar exponent H_*	0.9196	0.9114
Binned TV: 10 s versus 10,000 s	0.370	0.403
Largest TV among the six lags	0.370	0.441

Later rescaled laws concentrate more centrally, while their means also change. Overflow mass is zero to numerical precision in the displayed square.

Mean motion, centred spread, orientation and dependence can all contribute. The plot alone does not separate their physical causes.

Would separating mean displacement from centred shape clarify which assumption fails?

One-lag vector laws leave multi-time questions open

$$D_{\text{joint}}(\tau, \tau') = \frac{1}{2} \sum_{ij} |p_{\tau,ij} - p_{\tau',ij}|.$$

The grid has 64 interior bins per axis and overflow cells. The distance resolves changes at that bin scale; it is not a calibrated rejection test.

Even agreement of every one-lag vector distribution would leave the joint law at several times, and hence full process self-similarity, untested.

Which multi-time or phase-conditioned comparison would most help choose between candidate processes?

Compare transport magnitude before its directional structure

Two groupings

Alps, Pyrenees and Channel Coast use the defined geographic boxes.

Initial GNSS altitude h_0 defines:

Plains	$h_0 < 300$ m
Hills	$300 \leq h_0 < 800$ m
Low mountains	$800 \leq h_0 < 1500$ m
High mountains	$h_0 \geq 1500$ m

One flight-level estimator

$$m_f(\tau) = \frac{\sum_s n_{fs} m_{fs}}{\sum_s n_{fs}},$$

$$M_g(\tau) = \frac{1}{N_g} \sum_{f \in g} m_f(\tau).$$

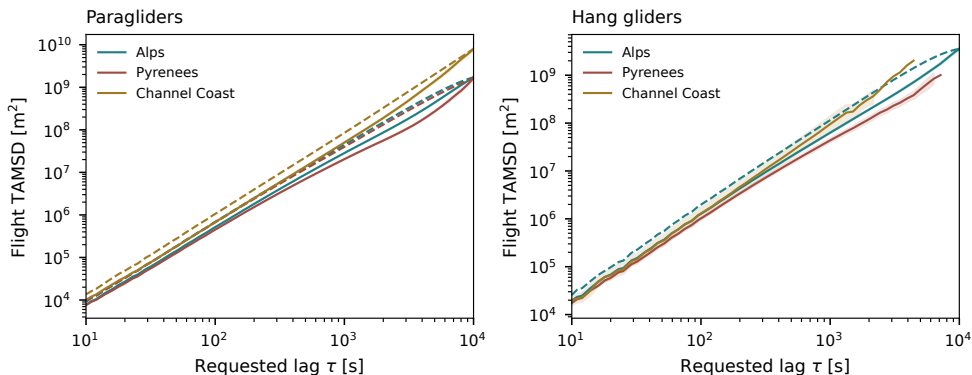
Pool supported origins inside each flight, then weight contributing flights equally.

Compare available segments with a fixed set supporting the largest lag.

How much of a group contrast concerns transport, and how much concerns which flights remain?

Native-grid TA-MSD; 155,085 para / 6,060 hang flights. Altitude groups include the full eligible archive.

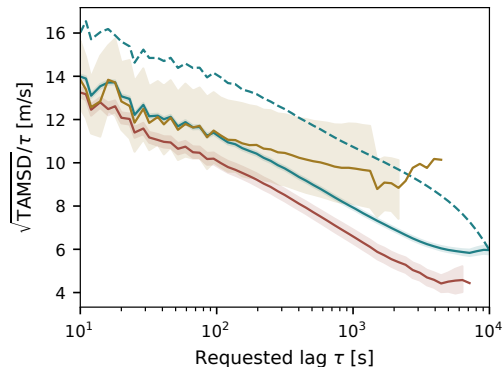
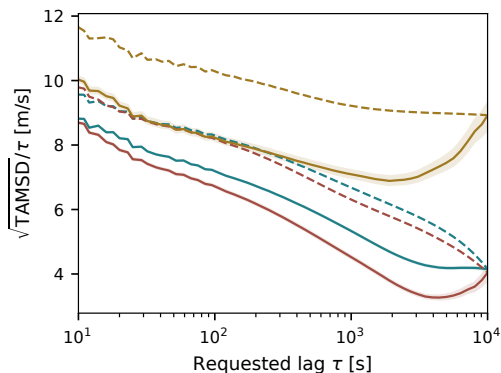
Regional TA-MSD: coastal flights travel farther at intermediate lags



Does the coastal ordering persist after restricting all lags to the same long segments?

Solid: available segments. Dashed: fixed long segments. Shading: pointwise 95% site-day bootstrap intervals for solid curves.

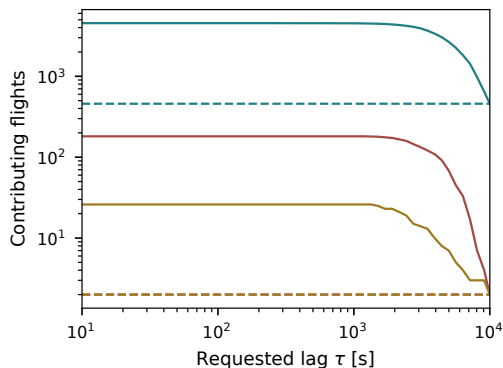
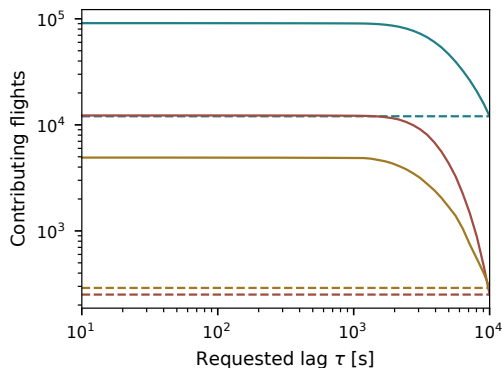
Fixing the population changes the apparent long-lag growth



Would the upward turn remain if short flights could contribute observed long-lag displacements?

The ordinate is RMS interval-mean ground speed, up to native-grid rounding. A horizontal curve corresponds to quadratic TAMSD growth.

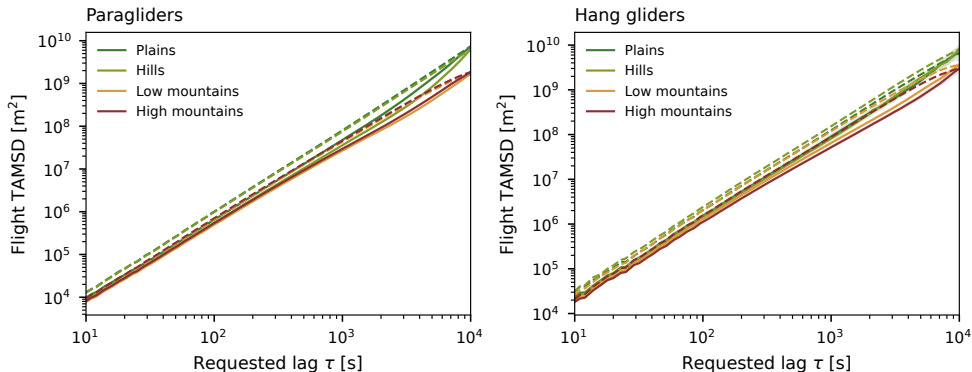
Regional support limits what the fixed control can establish



Which regional conclusions have enough long-flight support in both disciplines?

Fixed paraglider flights: 12,053, 251, 290.

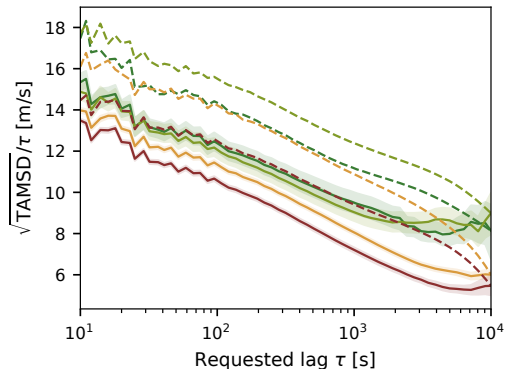
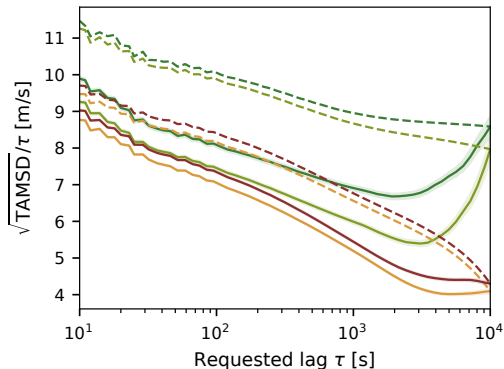
Initial-altitude groups combine transport and regional composition



Does an altitude-band contrast remain when region and contributing segments are held comparable?

Bands use the cleaned GNSS origin: Plains, Hills, Low mountains, High mountains. All eligible regions enter these curves.

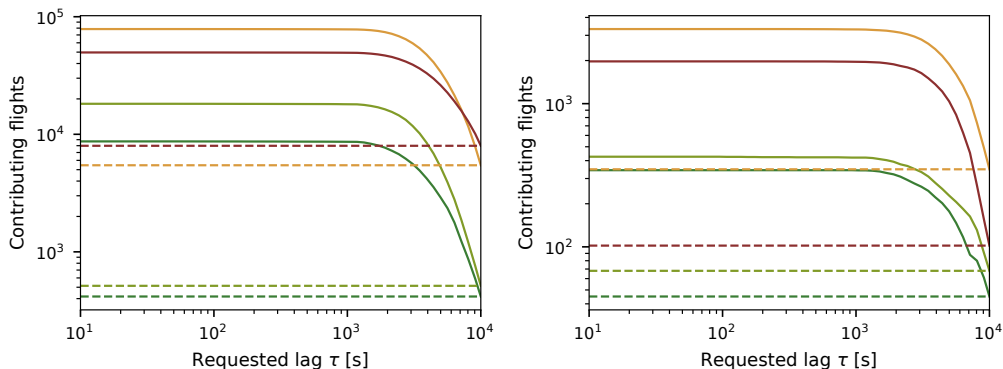
The late rise in Plains and Hills is sensitive to population selection



Which parts of the ordering survive the fixed long-segment comparison?

Solid: available segments; dashed: identical long segments at every lag. Shading is pointwise, not simultaneous.

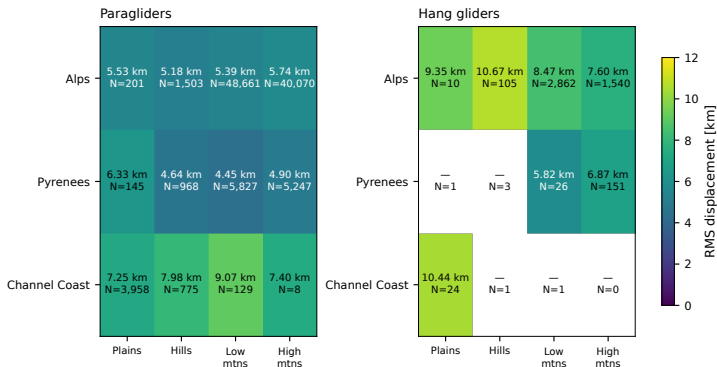
The altitude-band comparison also needs its flight counts



How much does the balance between altitude groups change as the lag increases?

Counts concern supported flights. The fixed-segment control retains membership; its origins still depend on lag.

Region and origin altitude are strongly associated in this sample



Can a pooled altitude contrast be interpreted as terrain when the same strata barely overlap?

RMS displacement at the requested 1049-s lag; available segments. Descriptive cells, with no calibrated intervals.

Regional plots use different clocks, weights and centring

Diagnostic	Clock	Weighting	Centring
Launch-time ratio	time since launch t	paired flights at each t	raw moments
Regional PCA	increment lag τ	eligible origins pooled	mean removed
Fixed joint law	increment lag τ	equal total flight weight	mean retained

These measurements differ in method as well as population. Compare these choices before attributing a difference to a physical mechanism.

Which apparent disagreement remains after comparing like populations and observables?

Equal east and north moments do not establish isotropy

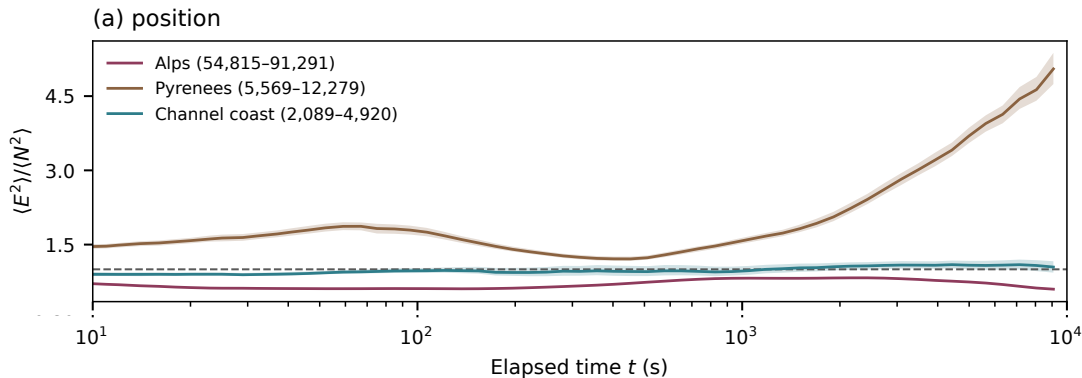
$$\rho_{EN} = 1, \quad \Sigma = \begin{pmatrix} 1 & 0.9 \\ 0.9 & 1 \end{pmatrix}, \quad \lambda_1/\lambda_2 = 19.$$

This centred Gaussian has balanced coordinate moments but strongly unequal principal variances.

A pooled mixture of east-directed and north-directed routes can also balance the ensemble while each flight remains directional.

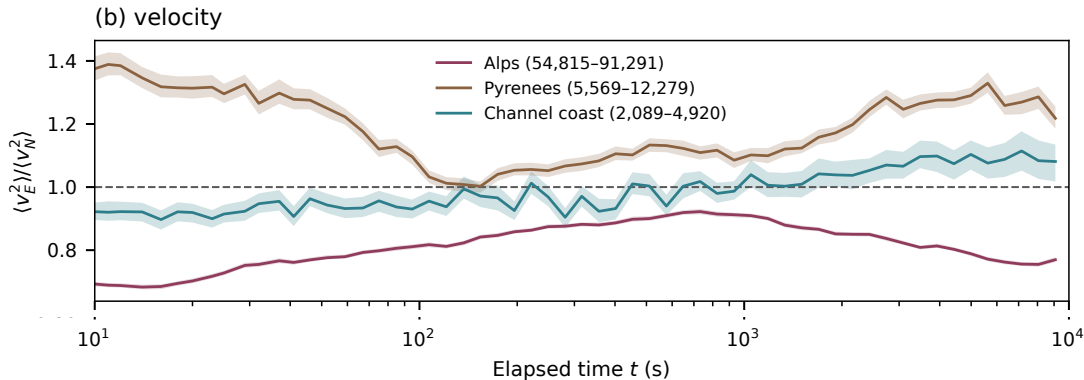
Are we observing mean drift, centred directional variability, or a mixture of individually oriented routes?

Regional position moments differ along east and north



How much imbalance reflects mean drift, route orientation or changing support?

Regional velocity moments show a related directional contrast



Which position and velocity contrasts persist under matched conditions?

Regional uncertainty bands do not match the groups

The ratios use paired east and north observations. A site–day bootstrap with 200 draws gives pointwise 10–90% ranges; each plotted time needs at least 30 flights and 10 clusters.

Missing site/date keys are treated as singletons. Resampling these groups does not match the equipment populations on weather, task or duration.

Would a paired site–day comparison change the group difference and its uncertainty?

PCA describes preferred axes; wind needs independent information

A rotation preserves distance

$$\Sigma = U\Lambda U^\top, \quad |U^\top \mathbf{X}| = |\mathbf{X}|.$$

PCA decorrelates components at a chosen lag. It cannot repair radial collapse or remove temporal dependence.

Whitening changes lengths and only fixes second moments at the fitted scale.

Ground velocity has two unknown terms

$$\mathbf{v}_{\text{ground}} = \mathbf{v}_{\text{air}} + \mathbf{w}_h.$$

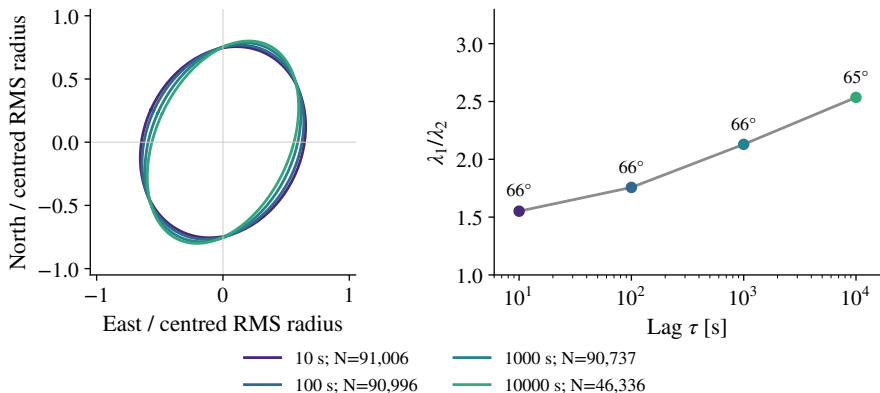
Terrain constrains headings; wind also changes position. Differentiation does not separate the two.

Different marginal exponents, or changing joint dependence, can prevent radial scaling.

Choose matched environments and an independent wind estimate before defining an “environment-free” model target.

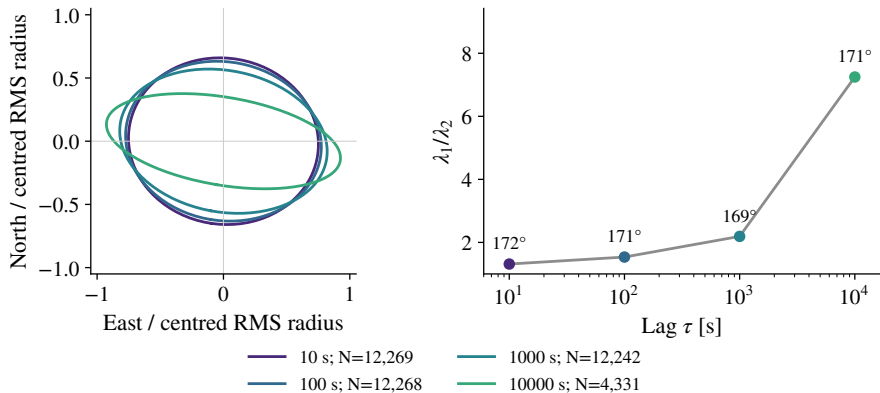
The following panels report regional PCA and its support. Anisotropy alone is compatible with a self-similar vector and a self-similar radius.

Alps: centred principal axes across four lags



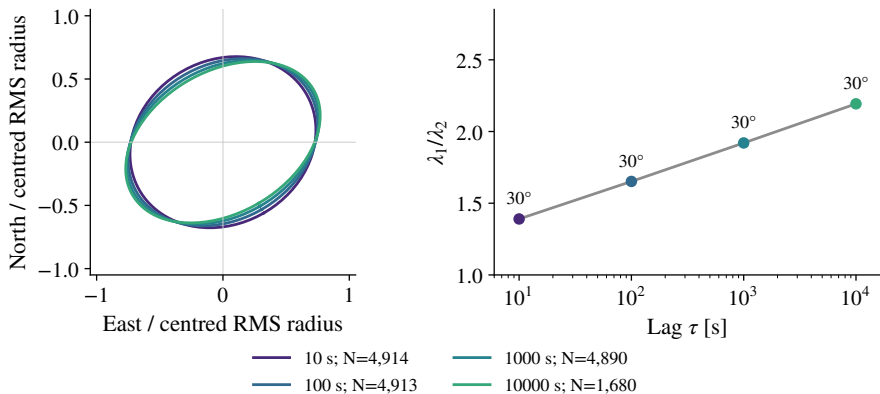
Does the axis remain stable as lag and contributing population change?

Pyrenees: centred principal axes across four lags



How much of the stronger 10,000-s elongation persists when flight identities are held fixed?

Channel Coast: centred principal axes across four lags



Does the axis remain stable as lag and contributing population change?

A terrain direction measured independently of the flights

NOAA ETOPO 2022

Same Alpine and Pyrenean take-off boxes.

One-arc-minute elevation sampling; retain the footprint above 1000 m.

Project cell centres to regional east/north coordinates. The area-weighted spatial covariance gives the main axis.

Definition sensitivity

Thresholds: 500, 1000, 1500 m.

Alps: 42.1–47.9°.

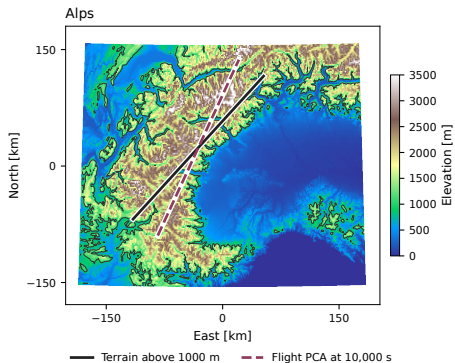
Pyrenees: 169.7–175.4°.

These spans are not confidence intervals; the axis describes an area, not a traced crest.

Would local axes along the actual tracks change the regional comparison?

[NOAA NCEI, ETOPO 2022](#). Both elevation subsets, requests, code and hashes are archived. Angles are counterclockwise from east, modulo 180°.

Alps: shared sector, with an approximately 18-degree offset



Terrain: 47.6° .

Flight PCA: 65.2° .

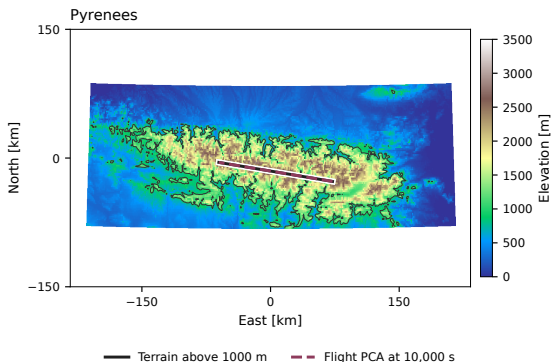
Separation: 17.6° .

46,336 flights support 10,000 s.

How much of the offset reflects Alpine curvature, box truncation or route selection?

Black: highland-footprint axis. Dashed colour: current flight PCA, shifted to the same centre. Line lengths have no physical meaning.

Pyrenees: the long-lag spread follows the highland axis



Terrain: 170.5° .

Flight PCA: 170.5° .

Separation: less than 1° .

4,331 flights support 10,000 s.

Does the alignment persist for fixed flights and for terrain sampled along their routes?

ETOPO footprint above 1000 m; threshold sensitivity: 169.7 – 175.4° . Decimal agreement does not imply this physical precision.

Channel Coast: the reference height follows the PCA windows

Same population as the 10,000-s PCA

- 1,680 paraglider flights.
- 2,013 nonoverlapping windows, wholly inside retained segments.
- Mean cleaned GNSS altitude along each window; equal window weights.

Mean: 1,006 m

Equal flight weights: 980 m.

Middle half of window means:
854 m–1,160 m.

Approximate altitude above sea level;
recorder datums are not harmonised.

Is one representative height sufficient, or should wind be matched to the altitude and location of each flight window?

Trapezoidal integration on the same 10-s segment grids. The target describes the supported long-lag windows; it is not take-off altitude or height above the terrain.

ERA5 wind is interpolated to the measured flight altitude

2016–2025, hourly; nine cells in the coastal box; 1000/925/850/700 hPa.

$$H = \Phi / g_0, \quad h = \frac{R_{\oplus} H}{R_{\oplus} - H}, \quad \theta_w = \text{atan2}(\langle v(h) \rangle, \langle u(h) \rangle).$$

Convert geopotential to each hour's level heights. Interpolate **E/N wind components** to 1,006 m, between above-ground bounds.

Main comparison: all year, all hours vs. all year, 09–17 UTC.

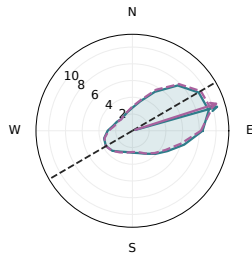
April–September is a separate seasonal check. Winds at 10/100 m AGL provide lower-level controls.

How much of the surface-to-flight-height contrast reflects vertical shear, season and the flight conditions represented?

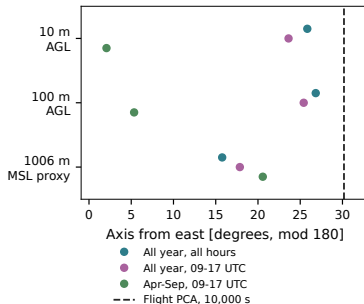
[Copernicus/ECMWF ERA5 pressure levels](#), pinned [Earthmover archive](#). Complete hourly extract: 32.22 MB compressed, preserved in the repository and on SSD.

At flight height, both annual references remain offset from PCA

At mean flight altitude: 1006 m
Towards-sector frequency [%]



Height and time selection



Annual gaps at flight height: 14.4° for all hours, 12.3° for 09–17 UTC. Why is the seasonal contrast larger near the surface?

Angles counterclockwise from east. The rose shows where air moves towards at 1,006 m; the black line is the 30.2° centred PCA axis.

Height and time checks preserve a broad orientation resemblance

At 1,006 m	Wind axis	PCA gap	ρ_w
All year, all hours	15.8°	14.4°	0.43
All year, 09–17 UTC	17.9°	12.3°	0.44
Apr–Sep, all hours	16.0°	14.2°	0.35
Apr–Sep, 09–17 UTC	20.6°	9.6°	0.37

At the quartile heights, annual axes span 14.3–17.2° (all hours) and 16.4–19.3° (daytime), on shared support.

Does the remaining offset reflect flight selection and route choices, or change when weather is matched to actual flights?

$\rho_w = \|\langle \mathbf{w} \rangle\| / \langle \|\mathbf{w}\| \rangle$ measures cancellation. Height ranges are sensitivities, not confidence intervals. The main estimates retain all selected hours.

What the environmental comparisons establish

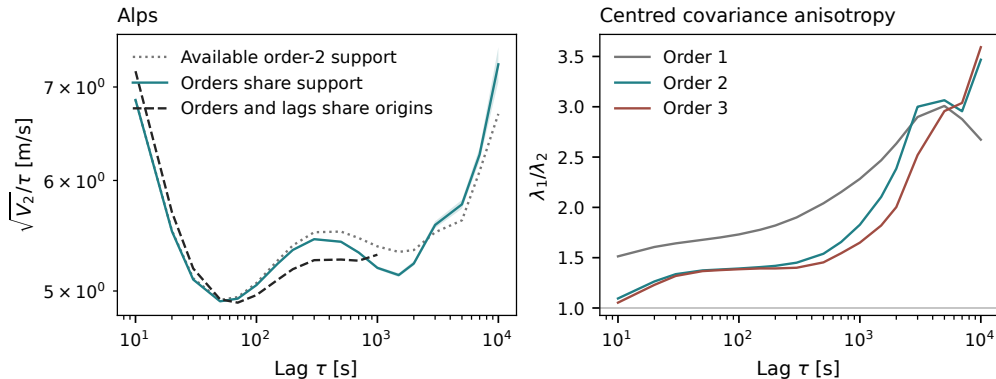
Region	Conclusion supported by this comparison
Pyrenees	Close terrain–flight alignment; a geographical association worth testing along tracks.
Alps	Same broad sector, with an approximately 18-degree offset from the mean highland axis.
Channel Coast	Wind at mean flight altitude lies about 12–14 degrees from PCA in both annual time selections; seasonal checks retain a broad resemblance.

A common constant wind shifts the mean displacement by $\tau\mathbf{w}$; centring removes it. Variable winds and route responses can still shape covariance.

Match flight times, heights and routes to wind and local terrain before attributing the ellipse or equipment contrasts to a cause.

Independent regional references constrain an interpretation. They do not separate wind, terrain, route choice and long-flight selection.

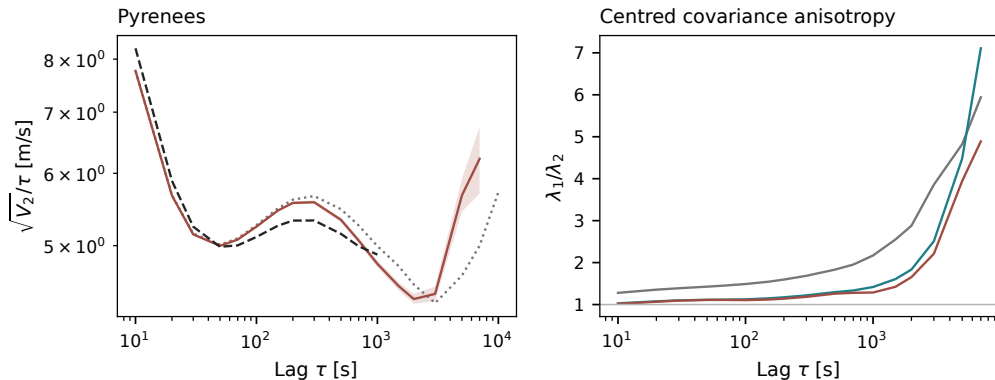
Alps: velocity changes retain a directional structure



Which part of the lag dependence persists with common origins and after changing difference order?

Equal flight weights. At 1000 s, matched orders retain 89,181 flights. Shading: pointwise 95% site-day intervals.

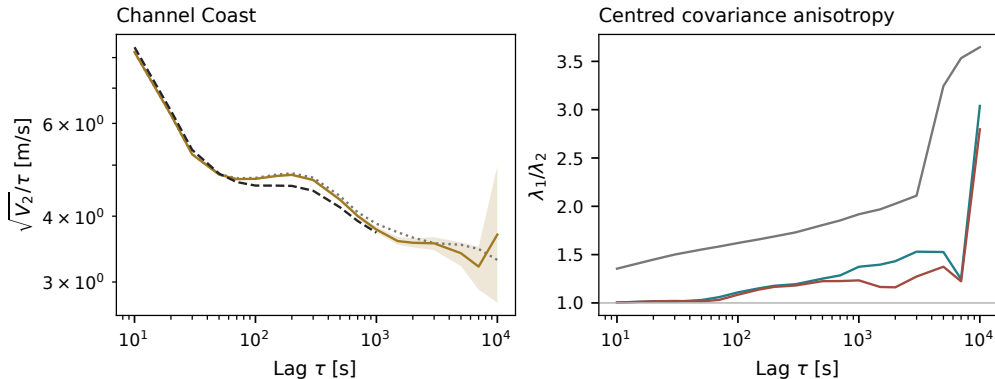
Pyrenees: the long-lag change coincides with falling support



How much of the late rise survives a comparison with fixed flights and origins?

At 1000 s: 12,020 matched flights; at 10,000 s: only 3. Cells below eight flights are omitted from plots.

Channel Coast: smaller changes at 1000 s, with residual anisotropy

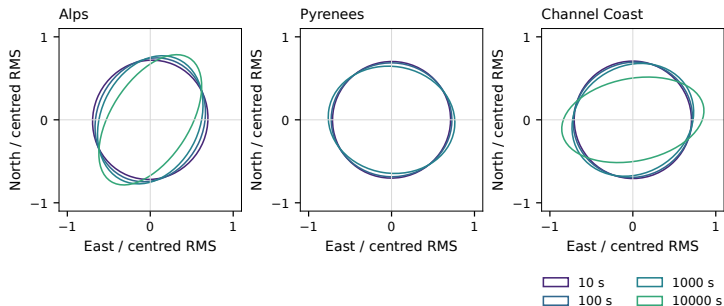


Could route geometry and varying forcing explain the contrast that survives constant-velocity cancellation?

At 1000 s: 4,679 matched flights, $D_2 = 3.77$ m/s and covariance ratio 1.37.

Second-difference ellipses retain orientation after drift cancellation

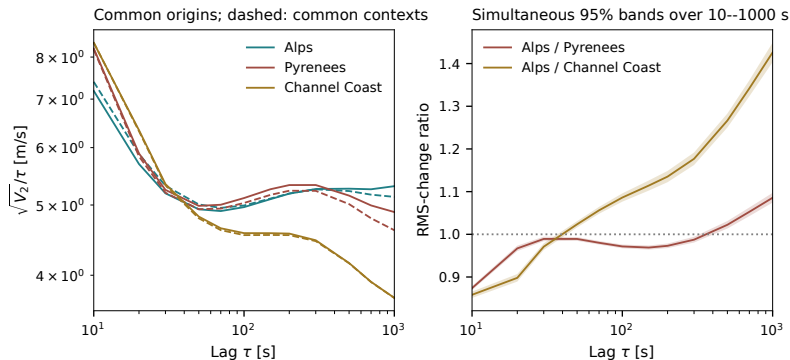
$$C_2 = \text{Cov}(A_2 \mathbf{r}), \quad V_2 = \text{tr } C_2 + |\langle A_2 \mathbf{r} \rangle|^2.$$



Do the resolved axes align with independent terrain and wind information?

Matched-order origins, equal flight weights; each ellipse is normalised by its own centred RMS.

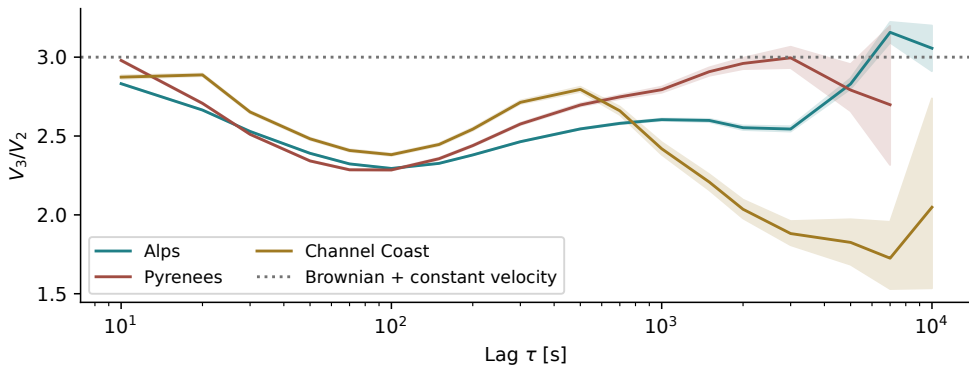
The regional contrast persists in a common-context control



Which part survives matching actual weather, sites and pilot identity?

Common origins through 1000 s. Dashed: 23 shared context strata, descriptive. Shading: simultaneous 95% bands for the unstandardised ratios.

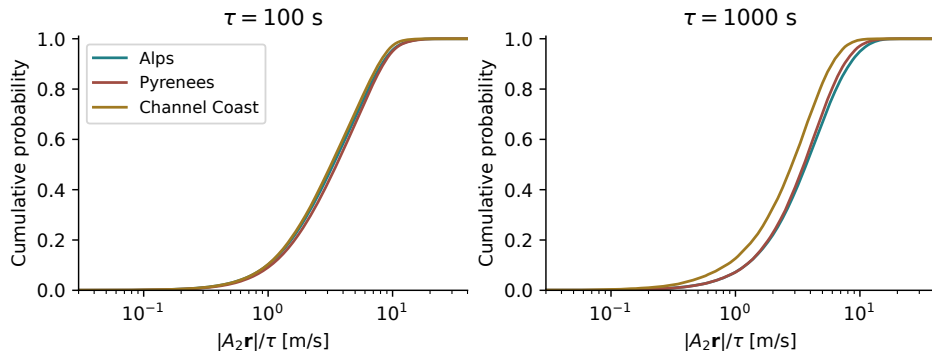
The ratio between orders changes with the lag



Which scale interval could satisfy both the slope and amplitude predictions of one model?

Matched origins and flight weights; pointwise 95% site-day intervals. Dotted: Brownian motion plus constant velocity.

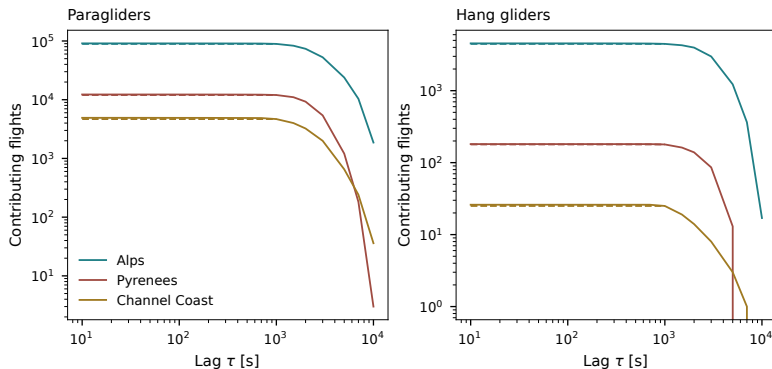
The distribution distinguishes frequent small changes from large events



Where on the trajectories do the largest contributions occur, and which verified flight phases contain them?

Equal-flight CDFs on matched-order support. Full report: all four display lags, zero and overflow mass, and quantile bin bounds.

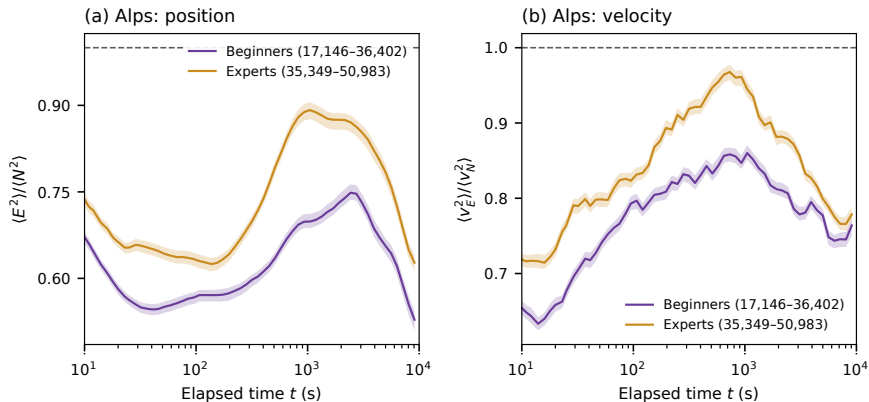
Order-three support limits the long-lag regional comparison



Which conclusions remain supported in both disciplines and across the same range of scales?

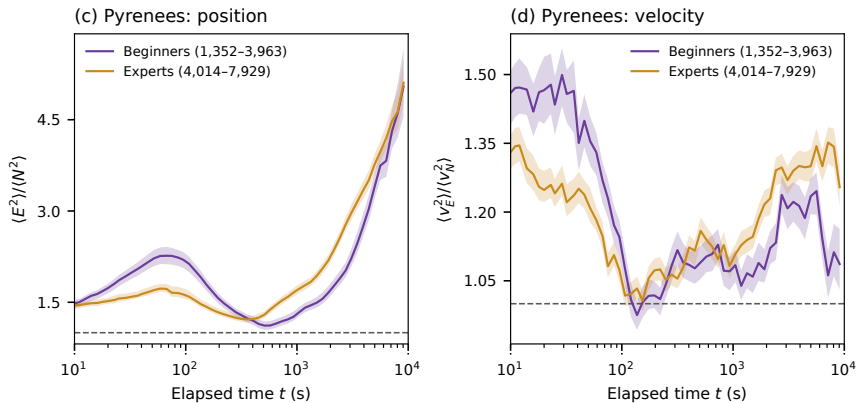
Solid: orders share 3τ support. Dashed: origins common through 1000 s. All eligible regional flights contribute where supported.

Alps: C/D/CCC has smaller coordinate-moment imbalance



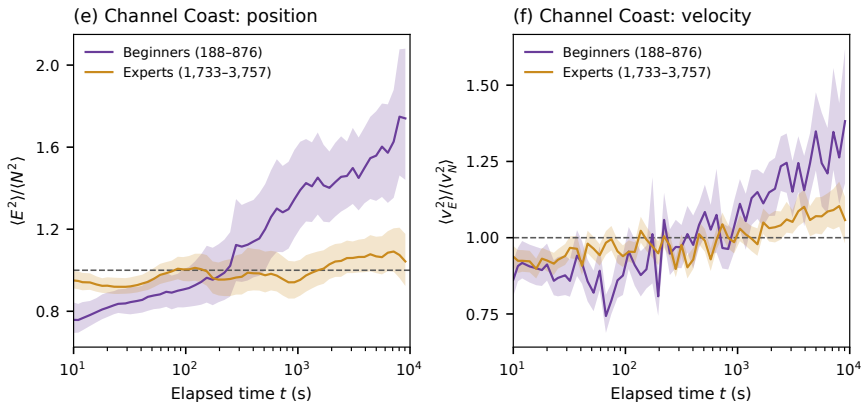
Does the ordering persist after matching site, day, task and duration?

Pyrenees: the ordering of the two groups reverses



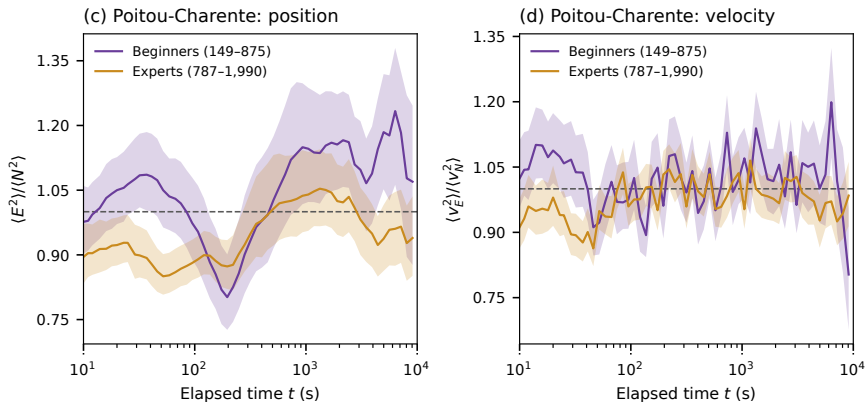
What changes between the early and later portions of the curves?

Channel Coast: the late position contrast is pronounced



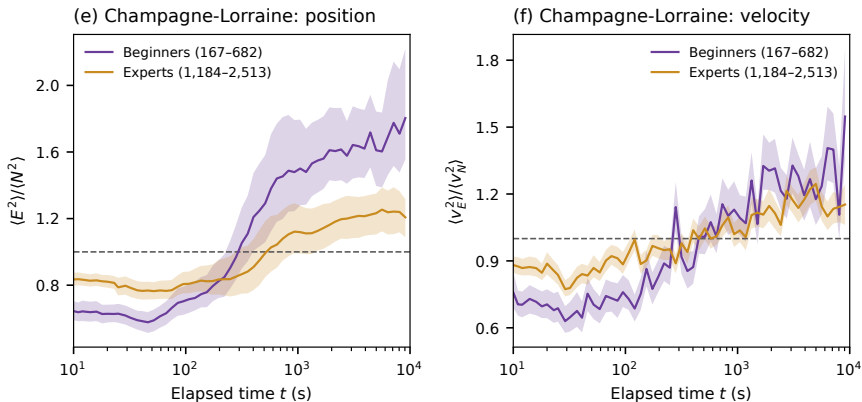
Can matching conditions separate directional route choice from wind response?

Poitou-Charente: the group curves cross



Does this low-relief launch box reproduce the coastal observation?

Champagne-Lorraine: a late contrast does not keep C/D/CCC at unity



Which parts of the coastal hypothesis transfer to this separate region?

Hypothesis to discuss

Equipment and pilot choices may moderate directional constraints associated with wind and terrain. Similar mountain profiles may reflect shared route constraints.

Evidence limiting that interpretation

The Pyrenean ordering reverses; class is an experience proxy; coastal terrain is not absent; late support differs; and ensemble balance does not establish individual isotropy or performance.

What observation would distinguish this hypothesis from different sites, tasks, weather or route mixtures?

Define the target before calling it resistance to wind

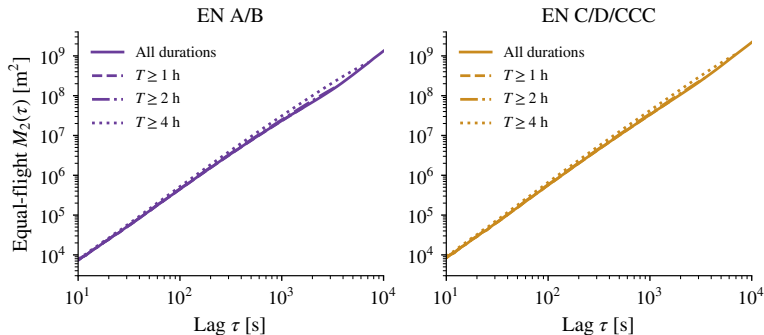
$$\mathbf{v}_{\text{ground}} = \mathbf{v}_{\text{air}} + \mathbf{w}_h.$$

Upwind progress, air-relative performance, choice of route and geographic moment balance are different outcomes.

A first comparison would match overlapping site-day, duration and task support; then separate mean drift, centred variability and individual route orientation. Regional references show terrain alignment and a remaining coastal wind-PCA offset at flight altitude; local track-matched observations are still needed.

Which target and matched comparison should be the first priority?

Long flights occur in both equipment groups



For $T \geq 4$ h: **7,415 A/B flights** and **27,815 C/D/CCC flights**.

Across 10–10,000 s, all-duration slopes are **1.728** and **1.788**.

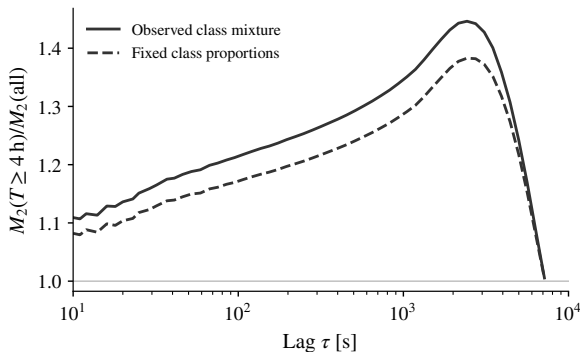
Equal-flight MSD; groups are nested. Threshold curves stop at $\tau = T_0/2$; at least 30 flights per lag. [All counts and common-range slopes](#).

Duration and equipment on the common 10–1695 s interval

Experience proxy	Duration	Flights	b	RMS residual (dex)
Beginners	All	54,292	1.755	0.022
Beginners	$T \geq 1$ h	51,124	1.754	0.022
Beginners	$T \geq 2$ h	31,768	1.767	0.020
Beginners	$T \geq 4$ h	7,415	1.806	0.017
Experts	All	94,283	1.803	0.017
Experts	$T \geq 1$ h	91,182	1.802	0.017
Experts	$T \geq 2$ h	69,087	1.811	0.016
Experts	$T \geq 4$ h	27,815	1.836	0.015

Counts are eligible flights, not independent windows; support falls with lag. RMS residual measures departures from a fitted line, not uncertainty. Threshold samples overlap.

Fixing equipment proportions removes part of the duration contrast



$$M_{T_0}^{\text{fixed}}(\tau) = \sum_c \pi_c M_{T_0,c}(\tau).$$

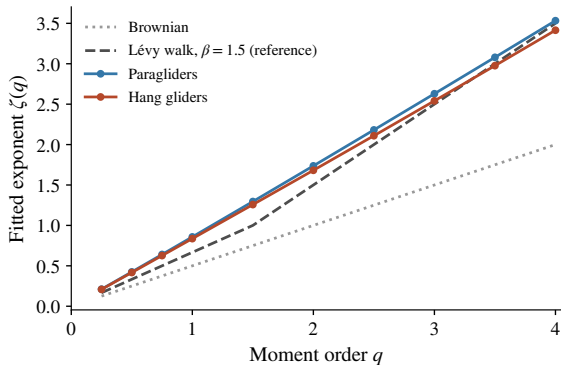
Fixed weights: $\pi_{A/B} = 0.365$,
 $\pi_{C/D/CCC} = 0.635$.

At $\tau = 1695$ s the ratio falls from 1.42 to 1.35. A within-class contrast remains.

Match site–day and task next. Convergence near 7200 s also reflects shorter flights leaving the reference.

Both curves use the same two known equipment groups. Equality at one lag does not establish independence from flight duration.

The moment spectrum challenges a specific Lévy-walk benchmark



$$M_q = \langle R^q \rangle \propto \tau^{\zeta(q)}.$$

For a constant-speed renewal walk with isotropic independent headings and $\psi(t) \sim t^{-1-\beta}$, $1 < \beta < 2$:

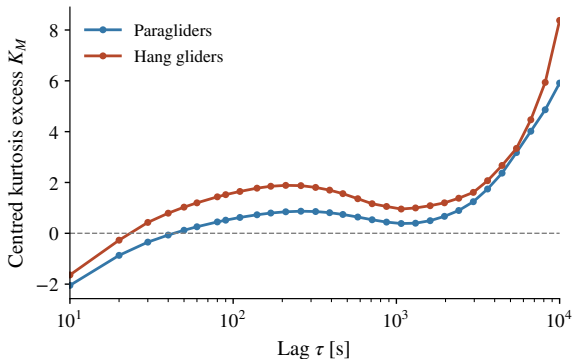
$$\zeta_{\text{LW}}(q) = \begin{cases} q/\beta, & q < \beta, \\ q + 1 - \beta, & q > \beta. \end{cases}$$

The empirical spectrum is nearly linear over $q = 0.25$ –4.

Can finite-record simulations resolve the predicted branch change under our sampling and selection?

Pooled windows. $\beta = 1.5$ is illustrative, not a fitted null; $q = \beta$ has a logarithmic correction. [Zaburdaev, Denisov & Klafter \(2015\)](#).

Non-Gaussian pooled increments can arise from Gaussian mixtures



$$\mathbf{Z} = \Sigma^{-1/2}(\mathbf{X} - \boldsymbol{\mu}),$$

$$K_M = \langle |\mathbf{Z}|^4 \rangle - 8.$$

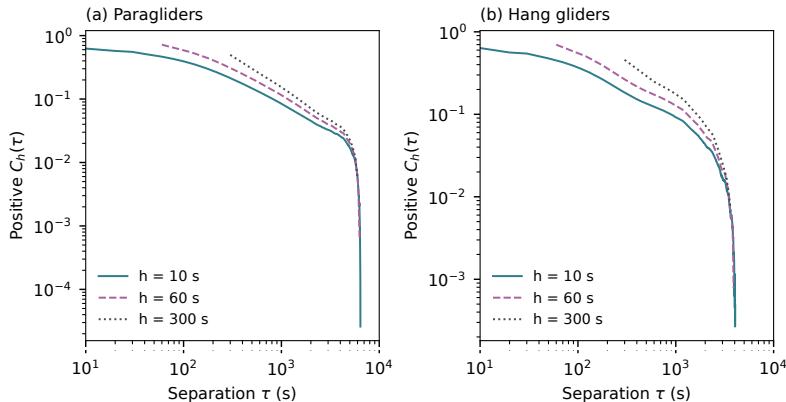
A nonsingular 2D Gaussian has $K_M = 0$, including anisotropic Gaussians.

The observed curve changes sign and rises at large lag, where support is weakest.

Test Gaussianity within matched environments or phases before rejecting conditional Gaussian dynamics.

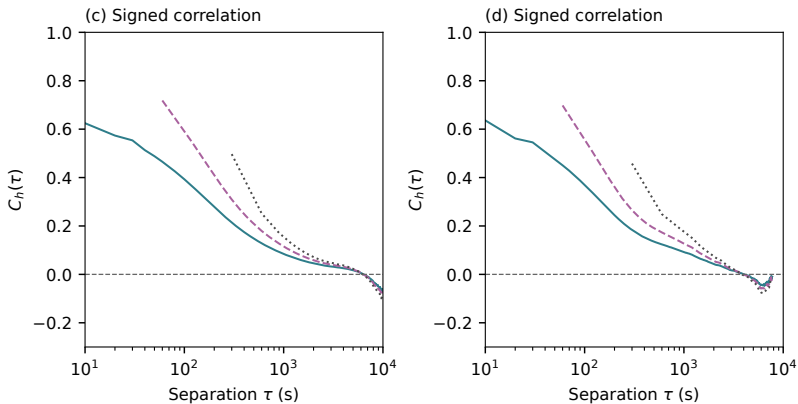
Centred, full-covariance Mardia excess; pooled windows. Positive excess does not identify a power-law tail. [Mardia \(1970\)](#).

Velocity memory remains visible after averaging over minutes



Do the curves support one decay law over the displayed scales?

Signed correlations retain the long-lag negative values



What share of the long-lag decrease could follow finite-record centring or route returns?

For the same stationary, centred velocity population,

$$M_{2,c}(\tau) = 2\sigma_v^2 \int_0^\tau (\tau - s)C(s) ds.$$

With $C(s) = e^{-s/t_p}$,

$$M_{2,c} = 2\sigma_v^2 t_p [\tau - t_p(1 - e^{-\tau/t_p})].$$

Ballistic at short times, diffusive at long times.

[Taylor \(1922\)](#), pp. 207–208. This covariance identity cannot be applied directly to differently weighted heterogeneous means.

Which models are challenged, and which remain open?

Candidate	Prediction to test	Present conclusion
Homogeneous Brownian	Linear MSD; independent increments	Challenged over the measured range.
Constant drift + Brownian	Drift-free $V_2 \propto \tau$	Simple version challenged; mixtures unresolved.
Homogeneous fBm	Gaussian law; common H_p	Challenged as one pooled law.
Exponential velocity memory	Ballistic–diffusive crossover	Remains a candidate.
Renewal Lévy walk, $1 < \beta < 2$	Two-branch moment spectrum	Finite-record comparison still needed.

“Challenged” means a descriptive mismatch. Calibrated statistical rejection is still to be done.

Each candidate needs its own parameters, environmental assumptions and observation protocol. A long-time diffusive limit has not been tested.

Phase-conditioned moments must include windows crossing transitions

$$M_q(\tau) = \sum_c w_c(\tau) M_q(\tau | c), \quad w_c = \frac{n_c}{\sum_{c'} n_{c'}}.$$

Here c partitions all admissible windows, including phase histories that cross boundaries. With equal flight weights, apply the identity within each flight before averaging.

The identity holds for any exhaustive labels. Recovering the global curve verifies accounting, not the physical accuracy of those labels.

Which episode durations and between-phase dependencies could distinguish renewal dynamics from persistent motion?

Three decisions for this meeting

1. Define the model target.

Ground-relative motion conditional on environment, or a residual after an independently justified wind correction?

2. Agree the first calibrated comparisons.

Fit Brownian controls, fBm, finite persistence and a specified Lévy walk; simulate the same record lengths, selection and estimators.

3. Choose the most informative next measurements.

Site-day uncertainty and route matching now; phase durations, increments and between-phase dependence after segmentation validation.

We have measured constraints on growth, shape, geometry and memory. The generating process and an intrinsic Hurst exponent remain unresolved.

Squared laws transform the same distributional evidence

$$Q_{Y^2}(p) = Q_Y(p)^2, \quad H_{Y^2,p} = 2H_{Y,p} \quad \text{for } Y \geq 0.$$

Squaring bin edges preserves their probability masses. A monotone transform also preserves the supremum distance between the correspondingly transformed CDFs.

If $f_Y(0^+) = c > 0$, then $f_{Y^2}(s) \sim c/(2\sqrt{s})$ near zero. This Jacobian effect says nothing about a heavy upper tail.

Which representation communicates the same result more clearly: displacement magnitude or its square?

Zaburdaev, Denisov & Klafter (2015), *Lévy walks*

Moment-spectrum benchmark

Mandelbrot & Van Ness (1968), *Fractional Brownian motions*

fBm construction

Taylor (1922), *Diffusion by continuous movements*

Velocity/displacement relation

Mardia (1970), *Measures of multivariate skewness and kurtosis*

Centred kurtosis

Efron (1979), *Bootstrap methods*

Resampling principle

Numerical evidence: current Chapter 3 reports, frozen in the accompanying data and asset manifests. Bibliographic access limitations are retained in the repository audit; citations do not validate the chosen sampling design or constitute calibrated rejection tests.

Fraiman et al., Cramér–Wold results

Didier & Pipiras, operator fractional Brownian motions

DHV classification guidance

DREAL Normandie, Le Pays de Caux

Joint laws and projections

Beyond scalar rescaling

Class is not a performance measure

Coastal cliffs and valleys

The cited pages support specific mathematical or observational cautions. They do not identify the causes of the regional contrasts measured here.